Licensed to: iChapters User

COMMERCIAL REAL ESTATE

ANALYSIS & INVESTMENTS

2e

GELTNER • MILLER
CLAYTON • EICHHOLTZ

Licensed to: iChapters User

Commercial Real Estate Analysis and Investments, Second Edition

David M. Geltner, Norman G. Miller, Jim Clayton, Piet Eichholtz

VP/Editorial Director:
Jack W. Calhoun

VP/Editor-in-Chief:
David Shaut

Executive Editor:
Scott Person

Associate Acquisitions Editor:
Sara Glassmeyer

Sr. Marketing Manager:
Mark Linton

Content Project Manager:
Erin Donohoe

Sr. Technology Project Manager:
Matt McKinney

Sr. Manufacturing Coordinator:
Charlene Taylor

Production House:
LEAP Publishing Services, Inc.

Compositor:
Interactive Composition Corporation

Printer:
West Group
Eagan, MN

Art Director:
Linda Helcher

Internal Designer:
Joe Pagliaro

Cover Designer:
C Miller Design

Cover Images:
Ed Pritchard/Stone/Getty Images

ALL RIGHTS RESERVED.
No part of this work covered by the copyright hereon may be reproduced or used in any form or by any means—graphic, electronic, or mechanical, including photocopying, recording, taping, Web distribution or information storage and retrieval systems, or in any other manner—without the written permission of the publisher.

Library of Congress Control Number:
2006908590

Printed in the United States of America
1 2 3 4 5 09 08 07 06

Student Edition:
ISBN-13: 978-0-324-38214-3
Student Edition: ISBN-10: 0-324-38214-0
Student Edition with CD
ISBN-13: 978-0-324-30548-7
Student Edition with CD
ISBN-10: 0-324-30548-6

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

Chapter Outline

1.1 Space Market
- 1.1.1 Segmentation of Space Markets: The Immobility of Real Estate
- 1.1.2 Supply, Demand, and Rent in the Space Market
- 1.1.3 Supply Is "Kinked"
- 1.1.4 Relation among Supply, Development Cost, and Rent
- 1.1.5 Forecasting the Future Direction of Rents
- 1.1.6 Is the Supply Function Rising, Level, or Falling?

1.2 Asset Market
- 1.2.1 Capital Markets
- 1.2.2 Pricing of Real Estate Assets
- 1.2.3 Is the Asset Market Segmented Like the Space Market?
- 1.2.4 Magnitude of Real Estate Assets in the Overall Capital Market

1.3 Chapter Summary

Chapter 1

Real Estate Space and Asset Markets

Learning Objectives

After reading this chapter, you should understand:

- How to apply the basic economic concepts of supply and demand to real estate markets.
- The difference between the real estate space and asset markets.
- The concept of market segmentation within the space market.
- Why the long-run supply function in the space market is "kinked" and what that means to future rents.
- The difference between, and relative magnitudes of, the public versus private asset markets, and the difference between equity and debt capital.
- What a cap rate is and what determines cap rate prevailing in real estate markets.

GEbrain.com

Markets are perhaps the most basic of all economic social phenomena. In essence, a **market** is a mechanism through which goods and services are voluntarily exchanged among different owners. In this chapter, we will present a basic introduction to the two major markets that are relevant for analyzing commercial real estate: the space market and the asset market.

1.1 SPACE MARKET

The **space market** is the market for the usage of (or right to use) real property (land and built space). This type of market is also often referred to as the real estate **usage market** or the **rental market**. On the **demand** side of this market are individuals, households, and firms that want to use space for either consumption or production purposes. For example, a student renting an apartment is using space for housing consumption. A law firm renting an office is using space for production. Both these types of users are on the demand side of the space market. On the **supply** side of the space market are real estate owners who rent space to tenants.

The price of the right to use space for a specified temporary period of time is commonly called the **rent**. It is usually quoted in annual terms, per square foot (SF), though other methods are also used (such as monthly per apartment). The rental price (determined by supply and demand in the

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

space market) thus gives a signal about the current value of built space and the current balance of supply and demand for that space. If usage demand grows and space supply remains constant, rents will tend to rise, and vice versa.

1.1.1 Segmentation of Space Markets: The Immobility of Real Estate

Users in the market for built space generally need a rather specific type of space in a rather specific location. A law firm needs an office building, not a restaurant or retail shop or warehouse, and it may need the building to be in downtown Cleveland, for example. If so, office space in downtown Detroit, or even office space in suburban Cleveland, will probably not satisfy the firm's needs.

The supply side of the space market is also location and type specific. Buildings cannot be moved. A vacant office building in downtown Detroit may be architecturally perfect for the Cleveland law firm, but it is not in the relevant location. Similarly, buildings generally cannot change their type, for example, from shop to office, or office to apartment. While some buildings have been converted from one usage type to another (for example, the warehouses that have been turned into ''loft'' apartments in New York and other large cities), such conversions are relatively rare and often require considerable construction expenditure.

Because both supply and demand are location and type specific, real estate space markets are highly **segmented**. That is, space markets tend to be local rather than national, and specialized around building usage categories. The market for warehouses in Dallas exists as a functioning market. A market for warehouses in the United States does not really exist as a single space market. This is in contrast to nationally ''integrated'' markets, such as that for gasoline, for example, or steel, or financial capital. These latter are homogeneous commodities that can be moved easily from place to place.

Because of space market segmentation, rental prices for physically similar space can differ widely from one location to another, or from one type of building to another, in virtually the same location. For example, office rents on new leases in Chicago's Loop can be $23 per square foot while space in similar office buildings in Midtown Manhattan is not available for less than $33/SF. Similarly, apartment buildings can provide their owners with annual rents of $7/SF in suburban Dallas when retail buildings there are yielding $13/SF on new leases.[1] The office buildings in Chicago cannot be picked up and moved to Manhattan to earn a higher rent, nor can the apartments in suburban Dallas readily be converted into shopping centers.

In real estate, the primary geographic units of space market segmentation are **metropolitan areas** (also known as MSAs, short for ''metropolitan statistical areas''). An MSA, encompassing a central city and its surrounding suburbs, tends to be relatively integrated economically, culturally, and socially. By definition, most points within a metropolitan area are more or less within automobile commuting distance of most other points in the area. The entire metropolitan area is largely built on the same economic base, although that base may be diverse in some areas. However, even within the metropolitan area important geographic submarkets exist. For example, the **downtown** (or **central business district—CBD**) is a different submarket from the suburbs, and individual suburbs will differ from one another. So even within a single metropolitan area considerable geographic segmentation of space markets exists.

In addition to geographical segmentation, real estate space markets are segmented by **property usage type**. The major types of space markets for rental property include office, retail, industrial, and multifamily residential. Other smaller and more specialized markets also exist, such as hotels, health facilities, self-storage facilities, golf courses, and a wide variety of others. Because each of these markets has different types of firms or individuals on the demand side, and different physical, locational, and architectural requirements on the supply side, the real estate industry serving these

[1]These were the actual average rents reported in late 1992 by the National Real Estate Index.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

various space markets also tends to be segmented. Individual real estate professionals tend to specialize in serving one market segment or another. Real estate firms, such as national brokerage businesses, that do encompass more than one market segment tend to be organized in divisions that focus on the various segments.

1.1.2 Supply, Demand, and Rent in the Space Market

CENGAGE brain.com

Let us "zoom in" on a specific, historic real estate space market and see how the economist's basic concepts of supply and demand and market equilibrium can help us to understand what goes on in a typical functioning space market. We will consider (a somewhat stylized version of) the market for class A office space in downtown Cincinnati, Ohio, during the late 1980s and early 1990s.[2]

From the 1970s to the 1980s, office employment grew substantially in downtown Cincinnati. From roughly 24,000 office workers in class A buildings in downtown Cincinnati in the 1970s, by the mid-1980s there were 30,000 such workers. This growth in office usage demand was due partly to national factors, such as a structural shift in the U.S. economy (out of manufacturing and into services) that increased the number of office jobs. The growth was also due to local factors, as several national firms relocated their headquarters from other cities to Cincinnati. The 30,000 workers in the mid-1980s were occupying some 5 million square feet (SF) of space in about a dozen office towers typically 20 to 30 stories tall which defined the skyline of the city. The average rent being charged in these buildings was about $16/SF (per year).

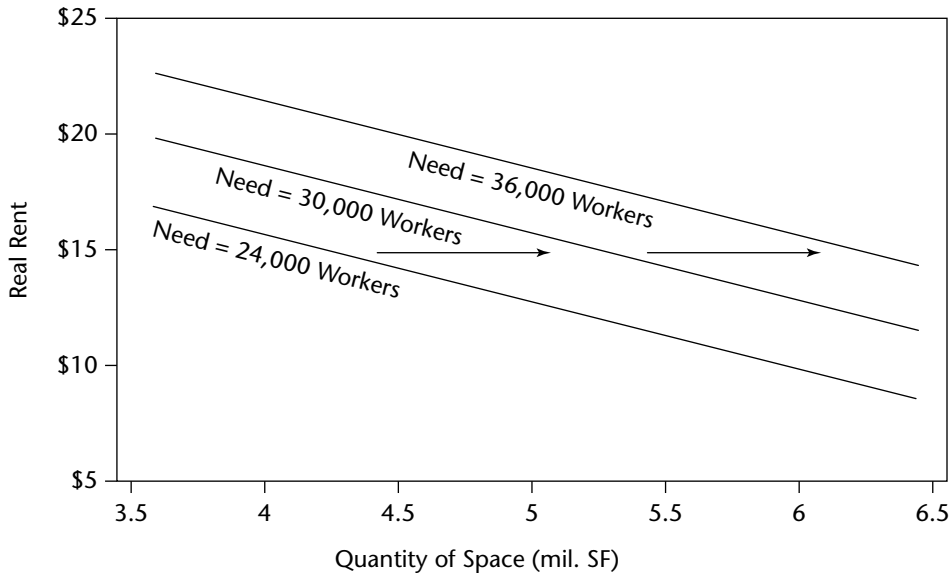
This growth in demand is pictured in Exhibit 1-1. The underlying source of the space demand was the need of office workers for space in which to work. As the number of office workers increased, this need grew. The need for office space also grew because technological change, such as the rise in the office use of personal computers and fax machines, made more space necessary per worker. The growth in class A office demand is represented in Exhibit 1-1 by the movement out to the right of the demand function line, for example, from a previous time when there were

[2]Real estate markets are often divided into classes, A, B, and C, to distinguish the physical and locational quality of the buildings, and therefore the levels of rent and types of tenants that are in the market on the demand side. Class A refers to the upper end of the market, where in the case of office buildings typical tenants would include prestigious law firms, corporate headquarters, and financial service firms. The particular historical period in this example is selected because of its educational value in illustrating a classic boom and bust cycle in the space market.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

Exhibit 1-1
Office Demand as a Function of Employment



24,000 workers to the time (by the mid-1980s) when there were 30,000 workers. Notice that if the underlying need for office workers in downtown Cincinnati increased further, to 36,000 workers, for example, then demand in the market would support an additional 1 million SF of space (a total of 6 million SF would be needed) at the same $16 rent.

The demand functions pictured in Exhibit 1-1 look essentially like the classical demand functions of economic theory—downward-sloping continuous lines that move out and to the right as demand grows. This is typical of demand functions in the space market. The supply side is a different story, however. The supply function in the real estate space market does not look like the upward-sloping continuous line depicted in the classical supply/demand diagrams. Instead, the possible supply function shapes shown in Exhibit 1-2 give a better representation of the supply side of most real estate space markets. This uniquely shaped supply function has important implications for understanding the functioning of the space market.

1.1.3 Supply Is "Kinked"

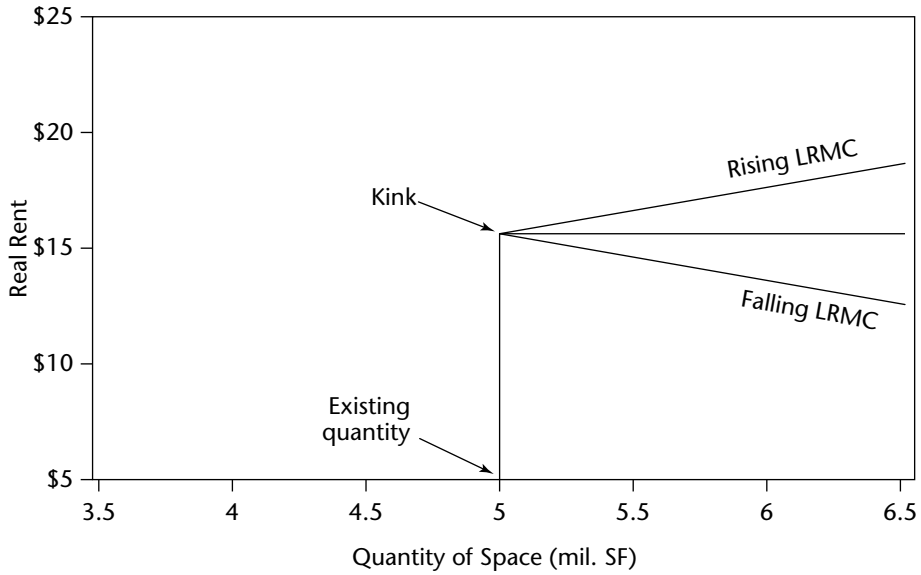
Economists often depict the typical real estate supply function as being "**kinked**"—that is, it is not continuous, but has a "corner" or break in it. The supply function starts out as a nearly vertical line at the current quantity of space supply in the market (in this case 5 million SF). This reflects the fact that the supply of office space is almost completely inelastic: if demand falls, office space cannot be reduced (at least in the short to medium term, even for a number of years). This is a consequence of the extreme longevity of built space. Indeed, compared to most other products, buildings last much longer. Rarely is a building torn down within less than 20 or 30 years from the time it is built, and it is much more common for buildings to endure 50 years or more (especially large-scale structures such as class A downtown office buildings). While some buildings can be converted to different uses, this is expensive and takes time. So at least for several years the market will maintain pretty much the quantity of supply it already has, in the case of a fall in demand.

The kink (or corner) in the supply function occurs at the current quantity of built space at a rent level that equates (on a capitalized present value basis) to the long-run marginal cost of supplying

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

Exhibit 1-2
Real Estate Space Market Supply Function



additional space to the market. Recall that the supply function for any competitively supplied product is simply the marginal cost function for producing additional increments of the product. In the case of built space, the marginal cost is the cost of developing new buildings, including the site acquisition cost as well as the construction cost and necessary profit for developers.

The level of rent that is just sufficient to stimulate profitable new development in the market is called the **replacement cost** level of rent, and this tends to be the long-run **equilibrium rent** in the market. If rents are said to be "below replacement cost" in a market, it means that developers cannot profitably undertake new development in that market. If rents rise above the replacement cost level, new development will be very profitable and will normally occur quickly in most U.S. cities, forcing rents back down to the long-run equilibrium level within a few years at the most.

1.1.4 Relation among Supply, Development Cost, and Rent

To understand the preceding point more clearly, let us return to our numerical example of the Cincinnati class A office market. Suppose that in the mid-1980s it would have cost $200/SF to develop a typical new class A office building in downtown Cincinnati, counting site acquisition cost as well as construction and other development costs, and including sufficient profit for the endeavor to be worthwhile to the developer. Thus, $200/SF was the marginal cost of adding additional office supply into the downtown market.

If we could sell new buildings for $200/SF, then additional development would be feasible. But the price at which one can sell buildings depends on what investors are willing to pay for them. This in turn depends on how much income the investors can expect to obtain from the buildings in the future, and how much investors will pay for that income. It is helpful in this context to quote property prices in terms of how much investors are willing to pay per dollar of annual net rent they would initially receive from their investment. Suppose, for example, that in the mid-1980s investors were willing to pay $12.50 to purchase a class A Cincinnati office property for each dollar of current annual net rent the property could produce. Then if a building could charge $16/SF annual net rent for office space, and expect to keep that space rented, the building would be worth

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

$200/SF (as 16 × 12.5 = 200). Using typical real estate terminology, we would say office buildings were selling at an 8% **cap rate**, as the net rent, $16/SF, divided by the building value of $200/SF, equals 8%. In these circumstances, the net rent that equates (after dividing by the cap rate prevailing in the market) to the marginal cost of adding office supply into the Cincinnati market is $16/SF. So $16 would be the replacement cost rent level. According to Exhibit 1-2, this was indeed the case, as indicated by the kink point at the $16 rent level.

The rest of the supply function in Exhibit 1-2 consists of a line moving out to the right away from the kink point. This part of the line will exhibit one of three possibilities—rising, level, or falling. A rising supply line results when the development cost of new buildings is greater as more total stock of supply is added into the market. Roughly speaking, if it would cost more to develop the next office building than it did to develop the last one (in real terms, after subtracting general inflation, and including the cost of site acquisition and any necessary demolition), then the supply function is rising beyond the kink point.[3] If it would cost less, then the supply function is falling. If it would cost the same, then the function is level beyond the kink point.

1.1.5 Forecasting the Future Direction of Rents

One of the most basic and important things you must have in the real estate business is an intelligent idea about the future direction of rents in the space market. The shape of the supply function is important, for it fundamentally determines the level of real rents as demand changes over time. In general, the kink means that if a space market is currently in equilibrium, future growth in demand will result in very little (if any) real increase in rent levels in most markets in the long run.

On the other hand, the kink implies that declines in demand will cause relatively rapid and severe reductions in market rents due to the inelasticity of supply reduction. Such reductions may be expressed, particularly at first, not so much in a reduction in the rents landlords ask for, but in an increase in vacancies and in the offering of ''concessions'' to potential tenants, such as months of free rent up front, as an inducement to sign a long-term lease. The kink in the supply curve is one reason real estate space markets have often tended to be cyclical, with periods of excess supply followed by periods of tight markets. A modest example of this occurred in the Cincinnati office market of the 1980s and 1990s. In the late 1980s, developers in that market, apparently anticipating further growth in office demand, built an additional 1 million SF of new office space, expanding the supply from 5 million to 6 million SF. This change in supply is indicated in Exhibit 1-3 by the movement of the supply function to the right, from S_1 to S_2, assuming a flat long-run supply function. Had the demand continued to grow, from D_1 to D_2 for example, this expansion of supply would have been justified, and rent levels in the market would have remained at the long-run equilibrium level of $16/SF. However, as happened in many real estate markets in the late 1980s and early 1990s, the supply overshot the demand. The demand growth did not materialize. The market remained stuck essentially with the same demand function, D_1. Combined with the new supply function of S_2, this caused market rents to plummet to levels around $13/SF. Indeed, during the general economic recession of 1990–1991 and the subsequent corporate downsizing, demand for class A office space in downtown Cincinnati probably actually fell, at least temporarily, from its previous level, perhaps to the level indicated by D_0, resulting in market rents temporarily in the $10/SF range.

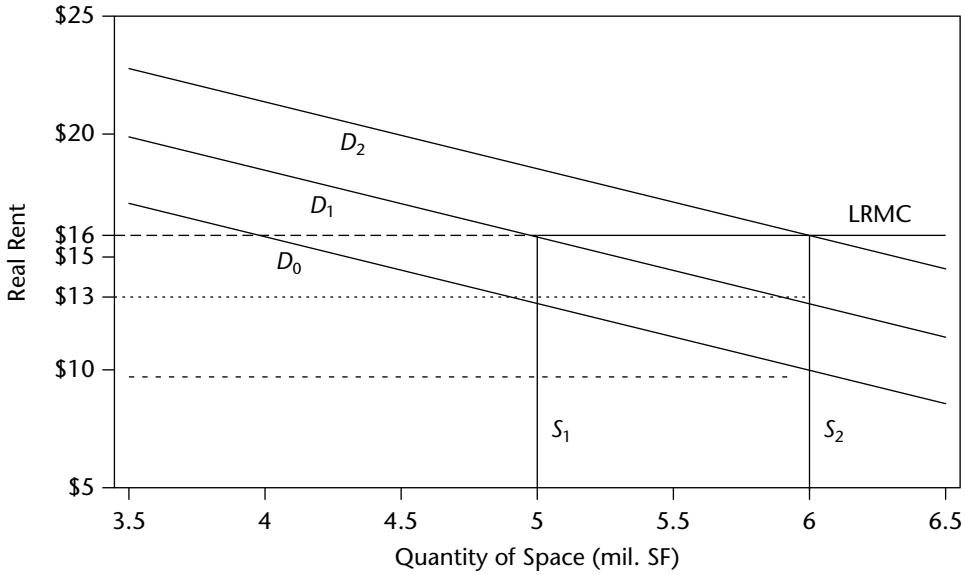
With demand at a level such as D_0 or D_1 and supply at a level such as S_2 in Exhibit 1-3, the market rents are below their long-run equilibrium level. That is, rents are below the level that can support new development, below replacement cost. At such a point, the space market is poised for a substantial surge in real rents if demand begins to grow again. Until rents rise back up to the long-run equilibrium level, the inelasticity of supply works in favor of landlords when demand is growing,

[3]In general, the term *real prices* refers to prices quoted in constant purchasing power dollars, after removing general inflation (or deflation).

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

Exhibit 1-3
Change in Supply and Demand and Rent over Time



enabling substantial hikes in real rents. (These may manifest at first as the filling up of vacancy and disappearance of widespread rent concessions in the market.) By the late 1990s, the class A office market in downtown Cincinnati was back up to long-run equilibrium rents.

1.1.6 Is the Supply Function Rising, Level, or Falling?

The principal cause of a rising supply function is land scarcity in the presence of growing usage demand. This results in increasing **location rent**, as space users are willing to pay more real dollars per year for use of the same location.[4] The real cost of building structures does not usually increase over time or from one site to another. However, if demand keeps growing in the face of fixed land supply, the real price of site acquisition will rise, and this will add to the cost of development.

One way that this occurs is that as more space is built in the market, the remaining supply of good buildable sites becomes scarcer. Construction will be more expensive on remaining unbuilt land sites that are hilly or swampy or less desirable for other reasons. Also, when new construction is undertaken on a site that already has a major existing structure on it, this greatly adds to the site acquisition cost. Not only must the existing structure be demolished, but the opportunity cost represented by the present value of the potential future income that the existing structure could otherwise generate is a major component of the site acquisition cost. For these reasons, it is possible for the supply curve to rise to the right of the kink point, as indicated by the upper line in Exhibit 1-2.

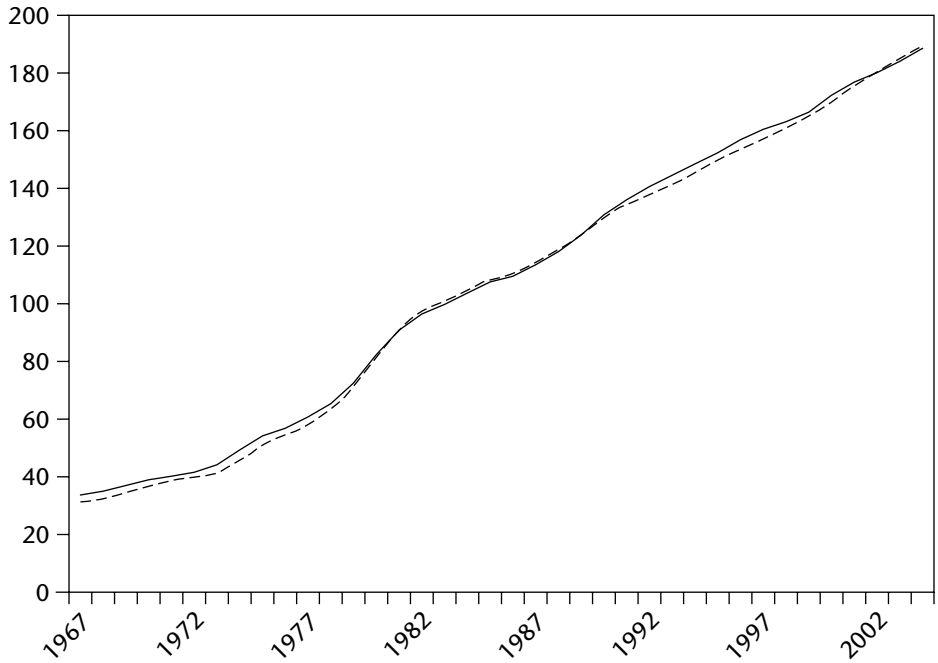
It is also possible, however, for the supply curve to decline to the right of the kink point. This can occur in areas where the location rent is declining over time in real terms. This occurs when a location is losing its relative **centrality**. For example, during the last half of the 20th century, the traditional CBDs of many metropolitan areas in the United States lost value relative to suburban locations. In general, the development of the automobile and telecommunications has reduced the relative value of central locations compared to peripheral ones. This has caused a reduction in the real level of long-run equilibrium rents for some types of commercial property in the United States.

[4]The term *location rent* will be discussed in more depth in Chapter 4.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

Exhibit 1-4
CPI General Inflation Index (solid line) and CPI Housing Component (dashed line): 1967–2004
1982–84=100



Evidence suggests that in "typical" or "average" space markets in the United States the long-run supply function has tended to be nearly level beyond the kink point, perhaps slightly rising or slightly falling. How do we know this? We can observe that rent levels in the economy have not generally tended to rise faster than inflation even as the total supply of built space has greatly expanded. For example, Exhibit 1-4 traces the history of the government's residential rent index compared with the general inflation index (the Consumer Price Index—CPI) since the mid-1960s. The solid line is the CPI, the number of current dollars it would take to purchase the same typical market basket of consumer goods and services at different points in time. The dashed line is the housing rent component of the CPI. Observe that the two curves are very nearly parallel, which indicates that residential rents or rent equivalents for owners, on average across the country, have risen at about the same rate as inflation. This means that real rents have not been increasing. This is in spite of the fact that the demand for housing over this period has grown, and supply has been added.

In judging the likely shape or change in level of the long-run supply curve in U.S. real estate markets, several general historical and geographical factors must be kept in mind. A fundamental consideration is that the United States is, relative to many other nations, a "land rich" country, or anyway perceives itself as such as part of our social and political culture. As a result, in the United States urban land use expansion is relatively lightly controlled. A fundamental historical factor that must also be kept in mind is that during recent history, massive transportation and telecommunications infrastructure investments occurred, including the development of the urban expressways that serve virtually all U.S. cities, as well as more recent developments in information and communication technology that tend to reduce the need for physical transportation and commuting. The result has opened up vast tracts of land for urbanization and exurban development by making all points readily accessible. As a result, the supply of land has not been at all fixed in most cities, but able to grow and expand with the demand. Large quantities of lightly used land still remain in and around many U.S. cities.

As noted, the effect of this physical expansion of urban land has been pronounced in the traditional CBDs. For example, in the downtowns of most U.S. cities, surface parking lots are not uncommon. These

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

represent a ready supply of unbuilt, lightly used land available for development. The historical evidence is that real rents in U.S. downtowns have not generally risen over the past half century, and indeed they have fallen in many cities. Exceptions to this rule are generally found only in severely space-constrained areas such as islands or peninsulas like Manhattan, Honolulu, and San Francisco.

In general, unless you have a good and specific reason to believe otherwise, it is probably safest to assume that the shape of the supply function is similar to the middle line portrayed in Exhibit 1-2 in most of the space markets you will probably be dealing in, at least in the United States. This may not be true in some other countries, however.

1.2 ASSET MARKET

While the space market is the most fundamental type of market relevant to the real estate business, the **asset market** is of equal importance. The real estate asset market is the market for the ownership of real estate assets. Real estate assets consist of real property, that is, land parcels and the buildings on them. Therefore, this market is often referred to as the **property market**.

From an economic perspective, such assets consist essentially of claims to future cash flows (e.g., the rents the buildings can generate for their owners). As such, these real estate assets can be compared to, and indeed compete in the capital market with, other forms of capital assets, such as stocks and bonds.[5] So the real estate asset market must be viewed as part of the larger **capital market**, the market for capital assets of all types.

1.2.1 Capital Markets

Broadly speaking, the capital markets can be divided into four categories according to whether they are public or private markets, and according to whether the assets traded are equity or debt. (See Exhibit 1-5.) **Public markets** are those in which small homogeneous units (or "shares") of ownership in assets trade in public exchanges in which many buyers and sellers are generally simultaneously participating in the market with price quotes available for all to observe. The stock market is the classic example of a public capital market. In such markets, transaction prices at which the units of the assets trade are typically reported publicly on a daily or more frequent basis. Public markets are characterized by a relatively high degree of **liquidity** in that it is generally possible to quickly sell units of the assets at or near the last quoted price. This liquidity is both a cause and effect of the fact that in public markets asset share prices can adjust rapidly to relevant news about their value. This ability of asset prices to respond quickly to relevant news is known as asset market **informational efficiency**.

In contrast, **private markets** are those in which the assets are traded in private transactions arranged between individual buyers and sellers who have "found" each other, often through the aid

Exhibit 1-5
Major Types of Capital Asset Markets and Investment Products

	Public Markets	Private Markets
Equity Assets	Stocks REITs Mutual funds	Real property Private equity Hedge funds
Debt Assets	Bonds MBS Money instruments	Bank loans Whole mortgages Venture debt & LBOs

[5]The net rents generated by a building are analogous in this respect to the dividends paid out by stocks or the interest paid out by bonds, and property competes for investors' capital against these other forms of investments.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

of brokers. In private markets, it is common (though not necessary) for **whole assets** (e.g., an entire company or property or oil well or mortgage) to be traded in a single transaction. The average size of the individual transactions therefore tends to be larger in the private markets.

Private markets generally are less liquid than public markets, as it takes longer for sellers to find buyers, and it is more difficult to ascertain a fair price for a given whole asset. Transaction costs (the cost of buying and selling) are typically higher in private asset markets (as a fraction of the value of the asset being traded). As a result of transaction and search costs, privately traded assets are generally traded less frequently than publicly traded assets. The fact that whole assets are traded in private deals between (typically) one buyer and one seller also has consequences for the nature of the asset price information that is available to the public.

In private asset markets, it will generally be more difficult (or expensive) to observe the prices at which assets are trading. Furthermore, as whole assets are generally unique (such as a piece of art or a parcel of land), it is difficult to know how relevant an observed transaction price is for judging the value of any other asset. This problem is compounded by the fact that the parties to each trade are also typically unique, so the observed price represents only the agreement of two parties, not a broad market consensus. The same assets are not usually sold repeatedly very frequently, which makes even the trend in prices over time difficult to observe. For all these reasons, asset values in private markets tend not to incorporate or reflect news and information as quickly as do prices in public asset markets. In other words, *private asset markets tend not to be as informationally efficient as public markets*.

Perhaps a quick example will be useful to make this point more concrete. Suppose you own an apartment complex near Los Angeles International Airport. You think the property is worth about $10 million, but of course you are not sure. Now you hear news that should affect the value of your property in a negative way. The FAA has announced new regulations that control the flight patterns approaching the airport, causing more noise and safety problems for your neighborhood. Logically, this reduces the value of your property, other things being equal, below the $10 million you previously thought it was worth. But by how much does it reduce the value? Is it now $9 million, or $9.9 million?

If another similar property near yours would sell, and you could observe that transaction price, this might give you some idea about how the value of your property has changed, but even this would not give a very clear or definitive picture. The other property is not just like yours, and the buyer or seller in that deal may be unique in some way, or may not have negotiated the way you would. And anyway, there may be no other similar property nearby that sells anytime soon. In short, you don't know how much to mark down your appraisal of your property's value as a result of the FAA news.

Contrast this situation with that of publicly traded assets. Suppose, for example, ownership of your apartment complex was divided into thousands of small homogeneous shares that were constantly being traded by numerous buyers and sellers at publicly quoted prices on the stock exchange. In that case, you would know very quickly and certainly the effect of the FAA news on the value of your shares, simply by observing the change in the share price.[6]

Now let's move from a comparison of the columns in Exhibit 1-5 to a comparison of the rows, to consider debt versus equity. Debt and equity are two different types of capital assets, both of which can be traded in either public or private types of asset markets.

Debt assets are essentially the rights to the future cash flow to be paid out by borrowers on loans they have taken out (e.g., interest payments and retirement of principal). The characteristic feature of debt assets is that they give their owners (e.g., the lenders or bondholders) a relatively "senior" or "preferred" claim for obtaining the cash which the **underlying asset** (the source of the cash flow) can generate. Also, debt cash flows are contractually specified, so the recipients know more precisely with relative certainty how much future cash flow they will be receiving, and when. Debt assets are therefore typically less risky than equity assets. Debt assets also typically have specified "maturity," that is, finite

[6]On the other hand, it may be easier for public market asset prices to *overreact* to news, resulting in a subsequent "correction" to the price.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

lifetimes after which there will be no further cash flow. For example, a bond might pay 8% of its stated (par) value in interest each year for 20 years, and then be retired by payment of the par value.

By contrast, **equity** assets are those that give their owners the "residual," or "subordinated," claim on the cash flows generated by the underlying asset (e.g., a company or a property). Because equity claims are the residual (the amount that is left over after senior claims, including those of any debt-holders), equity tends to be more risky than debt. But the equity owners also typically have more control over managing the underlying assets, and more ability to benefit from growth or "upside" potential in the cash flows generated by the underlying assets. Equity assets, such as stocks and real property, are typically infinite-lived. Infinite life means that owners of an equity asset must sell the asset in order to "cash out" or liquidate their holding. It also means that more of the present value of the asset derives from the expectation of cash flows in the distant future, where there is the most uncertainty. This is another reason that the value of equity assets tends to be more volatile than the value of debt assets.

Capital asset products based on real estate have been developed for, and are traded in, all four of these branches of the capital markets. For example, REITs (Real Estate Investment Trusts) offer publicly traded common stock shares in companies that essentially do nothing but own (and manage, and buy and sell) income-producing properties and mortgages.[7] Mortgage-backed securities (MBS) are publicly traded bondlike products that are based on underlying pools of mortgages, which are real-estate-based debt products. There are both residential MBSs and commercial MBSs (the latter known as CMBS). On the other hand, many mortgages, especially large commercial mortgages, are held privately as "whole loans," and these would be traded privately (if at all).

But the most fundamental form of real estate asset, the direct ownership of whole properties, is traded in a private market known as the property market, in which individual commercial buildings, single-family homes, and land parcels are bought and sold among parties who have often found each other through the services of real estate brokers. Ultimately, all the other forms of real-estate-based capital asset products are based on privately traded property as the underlying assets. That is to say, property assets produce the cash flow that is the ultimate source of any income or value for any real estate investment products or vehicles, including REITs and mortgages.

The demand side of the property market is made up of investors wanting to buy property. The supply side consists of other investors (or developers or owner users) who want to sell some or all of the property they currently own. Investors in the property market fall generally into two main types: individuals and institutions. The major types of institutions include REITs, pension funds, life insurance companies, and mutual funds or opportunity funds of various types. Private individuals invest in the property market both directly and through funds or syndications. In addition to U.S. institutions, many investors in the U.S. real estate asset market today are foreign individuals and institutions.

In a modern highly developed economy such as that of the United States, the capital markets are highly integrated and sophisticated, well-functioning arenas in which vast quantities of money and information are managed. The capital markets determine how the wealth and capital resources of the society are allocated, which in turn determines the future growth rate and economic production patterns in the economy. For example, if investors decide that the future earnings potential of, say, airplane manufacturers looks relatively good, while that of office buildings looks relatively bad, then airplane

[7]In fact, most REITs primarily own equity interests in real property, as opposed to mortgages. To be more specific, a REIT is a corporation or trust that has elected REIT tax status. As long as the REIT conforms to certain government requirements, the REIT is exempt from corporate income tax. These requirements essentially limit the REIT to real estate investment and development type activity and require that the REIT pay out 90% of its earnings each year as dividends to its stockholders. For the most part, REITs must hold for at least four years properties that they develop or purchase, which prevents merchant builders, such as home developers, from electing REIT status. The original idea behind the law that established REITs in 1960 was that REITs would be a vehicle through which small investors could invest in commercial real estate through the stock market, much the same way mutual funds allow small investors to hold diversified portfolios of individual stocks.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

stocks will rise and capital will flow to that industry, while office building prices will fall and developers will find it hard to obtain the funds necessary to build new office buildings. In this way, the capital markets govern the real physical allocation of the productive resources of the economy.

Because real estate makes up a large and important part of the overall capital market, the real estate components of the capital markets are quite important in this overall allocation process. In fact, the value of all real estate assets is comparable to that of the stock market in the United States.

1.2.2 Pricing of Real Estate Assets

As noted, the supply side of the property market consists of property owners wanting to sell or reduce their holdings of real estate assets, while the demand side of the market consists of other investors wanting to buy or increase their holdings of real estate assets. The balance between supply and demand determines the overall level of real estate asset values relative to other forms of physical capital in the country. Within this overall general valuation context, the specific values of individual properties or buildings is determined by the perceptions of potential investors regarding the level and riskiness of the cash flows that each individual property can generate in the future.

As individual properties and buildings differ greatly in size and magnitude, when speaking of property prices and values in general it is common to think in terms of property value *per dollar of current net rent or income*. This way, one can more easily compare prices across properties of different sizes. In fact, in real estate, especially in commercial property markets, a measure that is the *inverse* of this price/earnings multiple is most widely used to describe property prices and values. This measure, known as the **capitalization rate (cap rate** for short, also known as an **overall rate**—or **OAR)**, is simply the property operating earnings divided by the property asset price or value.[8] The cap rate is similar to a **current yield** (the amount of current income the investor receives per dollar of current value of the investment). So while the cap rate is an inverse measure of asset value (per dollar of earnings), it may be thought of as a direct measure of the current component of the return on the investment.

Thus, *property values can be represented as earnings (i.e., essentially, net rents) divided by the cap rate*.[9] The cap rate is determined by capital investment supply and demand in the asset market, based on three major factors.

1. *Opportunity Cost of Capital.* The prevailing interest rates and opportunities for earning returns in other forms of investments in the capital markets (e.g., stocks, bonds, money market instruments) are a major determinant of how much investors are willing to pay for any property per dollar of its current income. As noted, real estate assets are in competition for the investor's dollar in the capital market, competing not only with other properties but also with other forms of assets. When interest rates and yields on bonds and stocks are low, for example, investors cannot earn as much return from these alternative investments. At such times, investors will tend to be more eager to put their money into property, and they will not expect to earn as high a return from property as they otherwise would. This will raise the price they are willing to pay for the property per dollar of its current income, and reduce the cap rate.

[8]By convention, cap rates apply to property operating income and property value (i.e., as if there were no mortgage), not the equity investor's income or equity value if there is a mortgage.

[9]This is merely a conventional way to quote asset prices. It is not a statement of causality. That is, it would be too simplistic to suggest that property values are *caused by* the current income generated by the property and the current cap rate prevailing in the asset market for similar properties. As will be discussed in more depth in Chapters 9 and 10, property investors are more interested in total return (income and capital appreciation) than in current income alone, and such investors take a multi-year, long-run view of the potential and risks in the property's future income and value.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

2. *Growth Expectations.* Potential investors will be "forward-looking," considering the likely amount of growth (or decline) in the net rent they can expect the property to generate in the future. Of course, the net rent the property will be able to produce in the future depends on what the balance between supply and demand will be in the *space* market in the future. So to consider the growth question, investors must, in effect, try to forecast the future of the space market in which the property is situated. The greater the expected growth in future net rent, the more investors will be willing to pay for the property today per dollar of current net rent, and hence, the smaller the cap rate. (For example, if investors were willing to pay $15.00 instead of $12.50 per dollar of current net income, then the cap rate would be 6.67% instead of 8.00%, as 1/15 = 6.67%.)[10]

   As described previously, growth prospects for rents in a market depend on the current state of the market (e.g., are rents currently below the long-run equilibrium level?), how usage demand will change in the future (will it grow or stagnate?), and the shape of the long-run supply function in the market (is it flat or rising?). In addition to marketwide considerations, each individual property may have unique attributes that affect its own growth prospects, such as existing leases, need for capital improvements, and so forth.
3. *Risk.* If investors regard the future potential net income from the property as less risky and more certain (perhaps because they view the space market as relatively stable and easy to forecast), then they will be willing to pay more per dollar of current income for the property, also reducing the cap rate. Other things being equal, investors do not like risk and will pay more for less risky assets. Of course, investors' aversion to risk and their preferences for certain types of investments may change over time. For example, some investors may want certain types of assets or properties in order to diversify or balance their overall investment portfolios. The more investors prefer the type of asset that a given property is, the more they will pay for it and the lower will be its cap rate.

1.2.3 Is the Asset Market Segmented Like the Space Market?

Now that you are familiar with both the real estate space market and asset market and the role of the capital market in the latter, we may draw an important distinction between the real estate asset and space markets. While the building users who make up the demand side of the space market require the *physical* characteristics of real estate assets (i.e., they are looking for built space of a particular type in a particular location), this is not generally the case with the investors who make up the demand side of the real estate asset market. Asset market investors are seeking future cash flows—*financial* rather than *physical* assets. Fundamentally, they do not care how or from where the cash is generated. "Money is fungible," is how this point is often expressed. Cash generated from rents in Chicago warehouses is the same as cash generated from rents in Los Angeles shopping centers and as cash generated from General Motors Corporation stock dividends or bond coupons. This means that real estate asset markets are much more integrated than real estate space markets.

Recall that an important consequence of the lack of integration in the real estate space markets is that prices (in the form of rents) can differ greatly among cities or among types of property, even for properties that are physically similar. Similarly, the much greater degree of integration in the real estate asset markets has the opposite effect. Two properties with cash flows perceived as having similar risk and growth potential will sell at very similar cap rates, even if the properties are physically very different and their space rents out at very different rent levels.

[10]Note that this does not imply that investors are willing to settle for a lower *total* return, only that more of that return is expected to come in the form of appreciation or capital growth, and less in the form of initial income.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

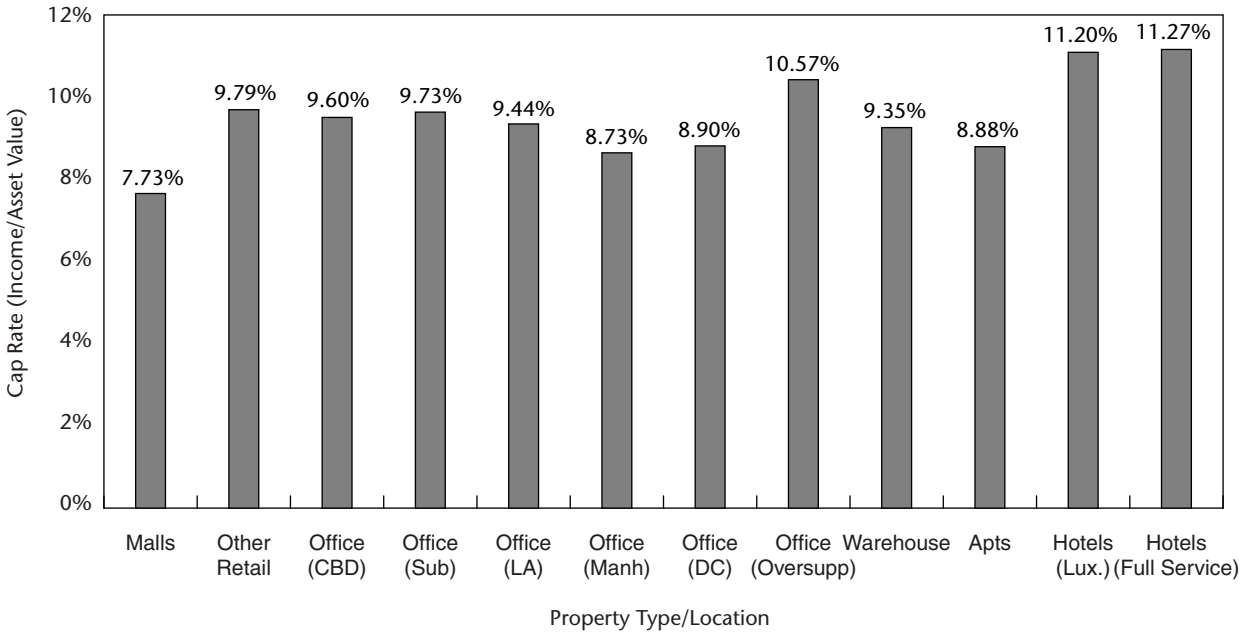


Exhibit 1-6 Cap Rates (OARs) for Commercial Property as of Third Quarter, 1994

To demonstrate this point more concretely, Exhibit 1-6 shows typical cap rates for a range of different types and locations of commercial property in 1994.[11] This is an interesting year to examine because at that time there were wide disparities in the level of risk perceived by investors in different space markets. Notice that the cap rates for office, retail, warehouse, and apartment properties were very similar (roughly around 9% or 10% per year), and this was true even across different cities. Where cap rates differed, it was not necessarily because rent levels differed or buildings were physically different, but rather because risks or growth prospects were perceived as being different.

This is seen in the cap rates reported for several types or locations of property in particular. For example, the "oversupplied office market" refers to office buildings in markets experiencing particularly high vacancy. Investors perceived such markets as being more risky than markets that were more in balance. So they required a higher expected return for investing in overbuilt markets, which equates to their being unwilling to pay as much per dollar of current income for such properties. In this case, we can see that the 10.57% cap rate indicates investors were generally not willing to pay more than about $9.46 per dollar of current income for office buildings in overbuilt markets, whereas in more typical CBD and suburban office markets they would pay over $10 per dollar of current income.

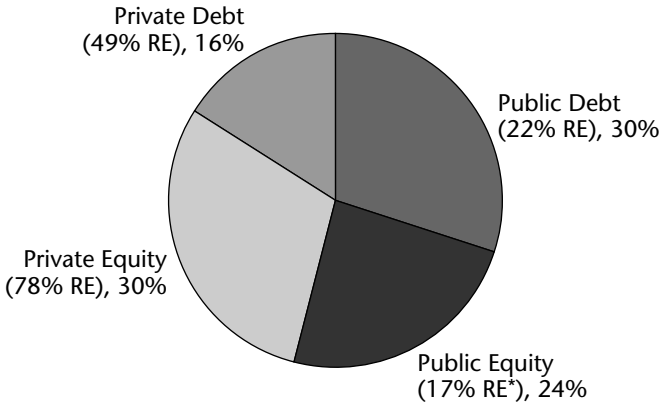
Hotel properties were perceived as being even more risky due to the lack of long-term leases from tenants and the volatile and highly competitive nature of the hotel business. In 1994, hotels were selling at cap rates in excess of 11% (price/earnings multiples less than 9). At the other extreme, regional shopping malls were perceived in 1994 as being relatively low risk, with relatively good growth prospects. Malls had average cap rates of only 7.73% in 1994, indicating property values of $12.94 per dollar of net income.

[11]Exhibit 1-6 is based on the *Korpacz Real Estate Investor Survey*, a widely used survey of professional investors and brokers published by the Korpacz Company, Frederick, Maryland. Other such data sources include *Real Capital Analytics* and *NCREIF*, among others.

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

Exhibit 1-7
U.S. Capital Market Sectors, a $70 Trillion Pie



*Corporate real estate owned by publicly traded firms, plus REITs.
Source: Authors' estimates based on Miles & Tolleson (1997), updated with FRB statistics.

1.2.4 Magnitude of Real Estate Assets in the Overall Capital Market

Exhibits 1-7 and 1-8 give some idea of the absolute and relative magnitude of real estate assets in the capital markets in the United States as of the early 2000s. Exhibit 1-7 shows the approximate proportion of the total market value of all U.S. assets found in each of the four sectors of the capital markets—public and private, debt and equity. Exhibit 1-7 also indicates the percentage of each of these sectors represented by real estate assets.[12] Altogether, real estate represents over one-third of the value of the investable capital assets in the United States. Real estate is particularly dominant in the private capital markets. Evidence suggests that these proportions are typical of most other advanced economies, with the real estate and private market shares being larger in less developed countries.

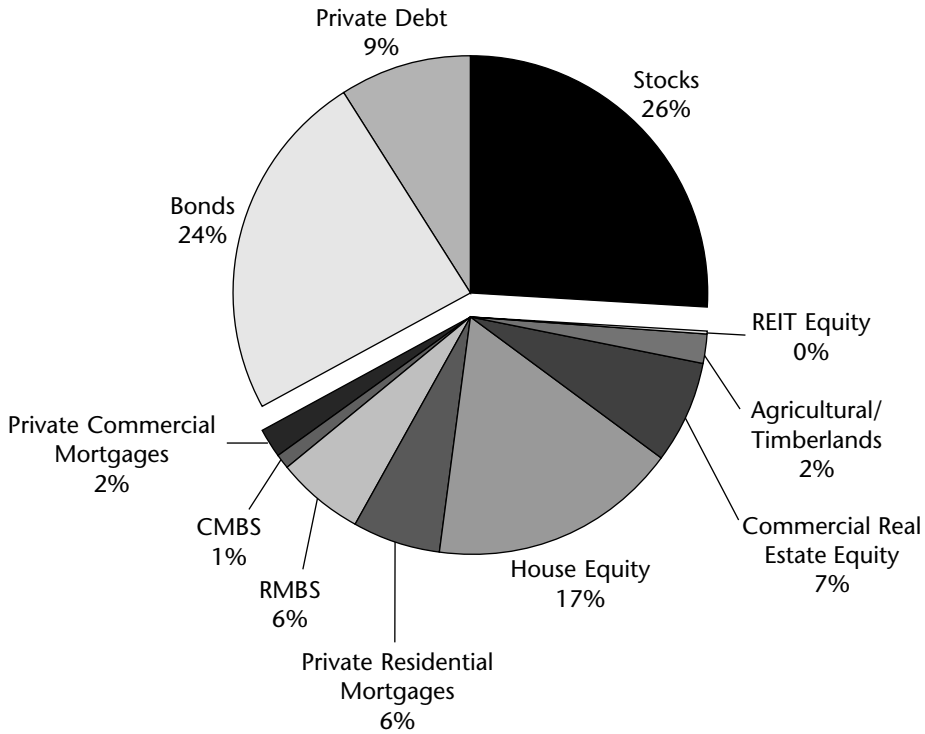
Exhibit 1-8 presents a more detailed breakdown of the capital asset "pie" by category of assets. The categories roughly between three o'clock and eight o'clock on the pie chart are real estate categories. The largest value categories include single-family housing equity (over $10 trillion in value), commercial real estate equity (almost $5 trillion including over $300 billion held by REITs and another $2–$3 trillion of corporate real estate held by publicly traded non-real-estate firms, which is categorized as part of the value of the stock market), residential mortgages (over $8 trillion, about half of which is included in publicly traded MBS products), and commercial mortgages (over $1 trillion of whole loans plus over $500 billion in publicly traded CMBS). Agricultural and timberlands represent over $1 trillion in assets. Altogether, real estate and real-estate-backed assets represented almost $30 trillion in the United States in the early 2000s, about two-thirds of that value in the form of single-family homes and the remainder commercial or income-generating property. This compared to an approximate stock market valuation at that time of only about $17 trillion.

[12]The figures in Exhibits 1-7 and 1-8 have been adjusted as far as possible to eliminate overlap. For example, a $100,000 house with an $80,000 mortgage would be counted as $20,000 of housing equity and $80,000 of residential mortgages (public debt if the mortgage was part of an MBS pool, or private debt if the mortgage was still held by the bank).

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

Exhibit 1-8
U.S. Investable Capital Market with Real Estate Components Broken Out



Source: Based on Miles & Tolleson (1997), updated with FRB statistics.

1.3 CHAPTER SUMMARY

This chapter has described the two major types of markets relevant to the analysis of commercial property: the space market and the asset market. The space market is the more fundamental of the two in the sense that it determines the cash flows property can generate, and these cash flows underlie any value the property asset can have. However, the asset market is of equal importance because it determines the valuation of property assets, and this in turn governs the flow of financial capital to real estate. Also, the asset market is most directly relevant to the analysis of investment in commercial property, which is the main subject of this book. In introducing these two real estate markets, this chapter has also presented a few of the most basic analytical tools and measures that are widely used in real estate investment practice, including such concepts as the kinked space market supply curve and the cap rate shortcut measure of real estate pricing in the asset market.

KEY TERMS

market
space market
usage market
rental market
demand
supply
rent
segmented markets
metropolitan areas (MSAs)
downtowns, CBDs
property usage type
kinked supply function
replacement cost
equilibrium rent
cap rate
location rent
centrality
asset market

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Licensed to: iChapters User

property market
capital market
public markets
liquidity
informational efficiency
private markets
whole assets
debt
underlying assets
equity
capitalization (cap) rate
overall rate (OAR)
current yield

STUDY QUESTIONS

Conceptual Questions

1.1. What is the real estate *space* market? What is the real estate *asset* market?

1.2. Draw and label a supply curve for real estate (income-property) space, and explain why it is "kinked."

1.3. What are typical causes of a rising long-run marginal cost function in the market for built space?

1.4. Why, or along what dimensions, are real estate space markets *segmented*? What is an important implication of space market segmentation?

1.5. Suppose that demand for built space in a local market is growing. Does this imply that rents will rise? Why or why not? Differentiate between rent changes in the short run and the long run. [Hint: For the long run, explain the conditions necessary for real rents to rise in a market.]

1.6. If demand for built space is declining, does this imply that real rents (that is, rents measured net of inflation) will fall in the market? Why or why not? As in the previous question, differentiate between short-run and long-run rent adjustments.

1.7. Why are cycles inevitable for real estate space markets? Do you think the nature of the cycles differs by property type?

1.8. What is the difference between a *private* and a *public* asset market? Why do public asset markets tend to be more *informationally efficient* than private asset markets?

1.9. Create a four-quadrant table showing on one dimension the two major types of capital asset products and on the other dimension the two major types of capital markets in which these assets are traded. [Hint: See Exhibit 1-5.] In which panel of this table would an individual commercial mortgage that is held by an insurance company fall? In which panel would ownership of a commercial building fall? In which panel would real estate investment trust stock (REIT) shares fall? In which panel would mortgage-backed securities (MBS) traded on the bond market fall?

1.10. What are the three major factors influencing property prices or cap rates (OARs) in the real estate asset market? How (i.e., in which direction) does each of these influence the rate?

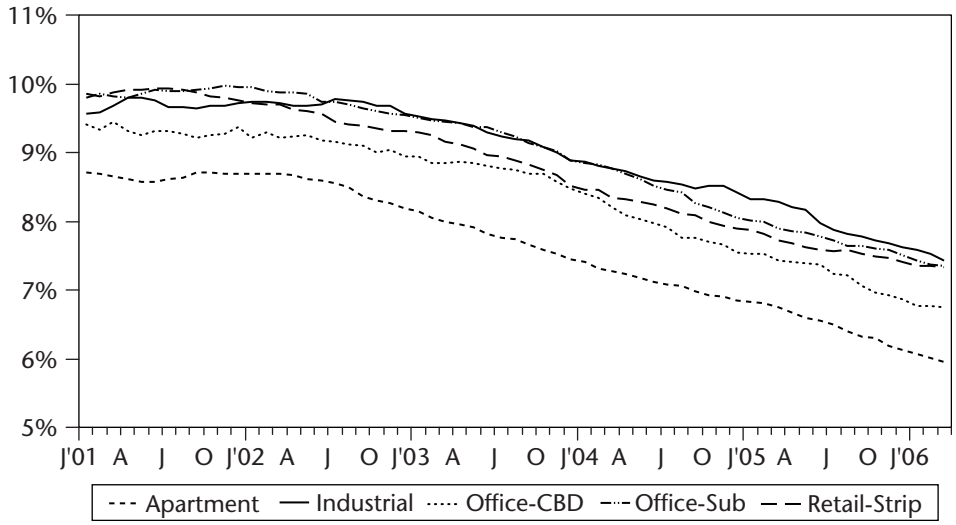
1.11. Exhibit 1-9 shows cap rates for office, industrial, retail, and apartment properties over the past five years.

a. Based on what you know about the three major factors upon which cap rates are based, what might explain why suburban office and industrial property have the highest cap rates, apartments the lowest cap rates, and CBD-office in the middle?

b. Cap rates fell quite dramatically over this period for all property types. How much of this decrease can be attributed to a decreasing opportunity cost of capital, as reflected in long-term (and risk-free) interest rates? To investigate this, visit the Web site http://www.federalreserve.gov/releases/h15/, and click on "Historical data." Then, go down to "Treasury constant maturities," and then click on the "Monthly" link for a 10-year. Create a

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.

Exhibit 1-9
Average Cap Rates for Major Property Types, Monthly, 2001–early 2006



Source: Real Capital Analytics.

graph of 10-year Treasury yields over this period to visually compare with the cap rates. Do cap rates appear to move one-to-one with Treasury yields? What else might be going on?

1.12. Approximately what percentage of the total market value of investable assets in the United States is made up of real estate or real-estate-based assets? [Hint: See Exhibit 1-8.]

1.13. **[REITs]** Go to the Web site of the National Association of Real Estate Investment Trusts (NAREIT) at http://www.nareit.org. Click on "Investing in REITs" on the left-hand side of the page and then "The REIT Story" under "Learn About REITs." Based on information contained in the article, answer the following questions:

a. What is a REIT? Be specific about any important criteria.

b. What are the three main types or categories of REITs? Approximately how many publicly traded REITs are there?

c. What are two main perceived benefits of investing in REIT shares?

Quantitative Problems

1.14. A property has a net rent (cash flow or net income) of $1.5m per year and is selling at a cap rate of 8.5%. What is the price of the property? Recalculate the price assuming a 9.75% cap rate and a 7.25% cap rate. [Note: This method of valuing income property, dividing the current net income by the cap rate, is known as direct capitalization, a widely used shortcut valuation method, which is discussed in greater depth in Chapter 10.]

1.15. Use the definition of the cap rate, equal to property income divided by property value, and the data on prevailing market cap rates for early 1994 presented in Exhibit 1-6 to estimate the value of (a) a typical high-quality downtown office building with a net operating income of $10 million per year and (b) a large apartment complex with a net operating income of $3.5 million.

1.16. Total development costs are $250/SF. Investors are willing to pay property prices of $11.76 per dollar of current rental income.

(a) What is the implied cap rate?

(b) What is the "replacement cost rent" in this market?

Copyright 2010 Cengage Learning, Inc. All Rights Reserved. May not be copied, scanned, or duplicated, in whole or in part.